

Foundations of Arithmetic Power Geometry XIII: The Spectral–Conductor Theorem and the Resolution of Open Problem 15.1

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Abstract

Arithmetic Power Geometry (APG) studies algebraic closure relations under exponent deformation and their analytic realization on compactified modular curves. Earlier APG volumes established entropy-governed local closure defects, information-geometric stabilization, integrated defect functionals, scale-corrected regularization, conditional APG–Arakelov height coupling, discrepancy–source compatibility, the Spectral–Green Bridge, source concentration, entropy-controlled spectral energy, spectral-gap stability for a canonical bounded Green-renormalized APG metric, and conductor-scale boundary dominance. This thirteenth paper completes the APG IX–XIII spectral-conductor route by proving Open Problem 15.1 under the canonical hypotheses inherited from APG IX, APG XI, and APG XII. Let

$$X = X_0(N)^*(\mathbb{C})$$

be the compactified complex modular curve of level N , let $\pi : X_0(N)^* \rightarrow E$ be the modular parametrization in the APG–Arakelov setting, and let Δ_{APG} be the positive APG Laplacian attached to the canonical bounded Green-renormalized metric. The main theorem proves

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C \deg(\pi) \log N.$$

The proof is a non-circular synthesis: APG IX bounds the mean-zero APG source by entropy, APG XI prevents spectral-gap collapse at polynomial conductor scale, and APG XII dominates the resulting entropy-conductor term by modular degree. The paper also records a conditional Fermat obstruction corollary: if a primitive Fermat triple forces a strict APG lower-energy gap exceeding the APG XIII conductor wall, then such a triple cannot exist. This final corollary is conditional and is not presented as a new unconditional proof of Fermat’s Last Theorem. The unconditional proof of Fermat’s Last Theorem remains the classical modularity and level-lowering proof.

Keywords: Arithmetic Power Geometry; spectral-conductor inequality; modular curves; Laplacian; spectral gap; Shannon entropy; Arakelov geometry; modular degree; Green functions; source concentration; Fermat obstruction.

1 Introduction

Arithmetic Power Geometry (APG) was introduced as a deformation-theoretic framework for studying how algebraic closure relations change when exponents are treated as deformation parameters rather than as fixed integers. The first APG volume formulated the Power-Deformation Space and proved that the leading local closure defect near the Euclidean target is governed by Shannon entropy (Akhtar, 2026a; Shannon, 1948; Cover and Thomas, 2006). APG II developed stabilized information geometry and coordinate stability in the spirit of information geometry (Akhtar, 2026b; Amari, 2016; Nielsen, 2020). APG III introduced the integrated closure-defect functional, while APG IV and APG V added scale correction, coordinate-dependent damping, and discrete prime-chain regularization (Akhtar, 2026c,d,e).

The later APG program moves toward the APG–Arakelov architecture. APG VI formulated a conditional height-coupling program relating APG deformation energy to the stable Faltings height of Frey-type elliptic curves through modular parametrization (Akhtar, 2026f; Frey, 1986; Faltings, 1983, 1984). APG VII established discrepancy–source compatibility and reduced the missing analytic-to-arithmetic projection step to cusp-width, vertical-component, and Green-energy boundary terms (Akhtar, 2026g; Arakelov, 1974; Soulé, 1992; Zhang, 1993). APG VIII proved the Spectral–Green Bridge on compactified modular curves, APG IX proved source concentration including cusp neighborhoods, APG X synthesized the entropy–spectral energy bridge, APG XI proved polynomial spectral-gap stability for the canonical bounded Green-renormalized APG metric, and APG XII isolated the conductor/modular-degree dominance estimate needed for the final synthesis (Akhtar, 2026h,i,j,k,l).

The present manuscript is APG XIII. Its purpose is deliberately narrow. It does not introduce a new APG invariant. It proves the final spectral-conductor theorem by chaining the estimates established in APG IX, APG XI, and APG XII. The target is Open Problem 15.1 from the APG 0 framework:

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C \deg(\pi) \log N. \quad (1)$$

This inequality is a spectral-arithmetic upper bound. It says that the mean-zero APG source fluctuation, amplified by the inverse spectral gap of the APG Laplacian, remains controlled by the modular degree and logarithmic conductor scale.

Classical arithmetic geometry studies such questions through heights, divisors, modular curves, Galois representations, and Arakelov intersection theory (Hartshorne, 1977; Lang, 1983, 2002; Bombieri and Gubler, 2006; Hindry and Silverman, 2000; Silverman, 1986, 1992, 2007; Katz and Mazur, 1985; Shimura, 1971; Diamond and Shurman, 2005). The surrounding technical landscape includes arithmetic surface theory, conductor-discriminant comparisons, modular-curve intersection estimates, discrete number-theoretic regularization, Diophantine approximation, and deformation-geometric analogies (Arakelov, 1974; Abbes and Ullmo, 1997; Borevich and Shafarevich, 1966; Deligne, 1971; Saito, 1988; Szpiro, 1981; Vojta, 1987; Maclagan and Sturmfels, 2015; Akhtar, 2026m). The proof of Fermat’s Last Theorem proceeds through the Frey curve, Ribet’s level lowering, and modularity lifting (Frey, 1986; Mazur, 1977; Serre, 1987; Ribet, 1990, 1994; Diamond, 1996; Taylor and Wiles, 1995; Wiles, 1995). APG does not replace that proof. Its contribution is a deformation-energy language and a spectral-conductor reduction within its

own canonical metric architecture.

2 Scope and Non-Claims

The paper proves the APG spectral-conductor theorem under the canonical APG IX–XIII hypotheses. It does not claim the following:

1. an unconditional new proof of Fermat’s Last Theorem;
2. a proof of the *abc* conjecture;
3. a proof of the Szpiro conjecture;
4. a proof of the Birch and Swinnerton-Dyer conjecture;
5. a proof of the full APG–Arakelov Projection Theorem beyond the stated conditional consequences;
6. spectral stability for arbitrary, unnormalized APG metrics.

The theorem is unconditional relative to the imported APG IX, APG XI, and APG XII results. The Fermat consequence at the end is explicitly conditional on a strict APG lower-gap hypothesis for primitive Fermat triples. This distinction is essential because APG VI and APG VII intentionally framed the APG-to-height and APG-to-Fermat bridge as a conditional projection program ([Akhtar, 2026f,g](#)).

3 Notation and Standing Setting

Let

$$X = X_0(N)^*(\mathbb{C}) \tag{2}$$

be the compactified complex modular curve of level N . Let

$$\pi : X_0(N)^* \longrightarrow E \tag{3}$$

be the modular parametrization into the elliptic curve E appearing in the APG–Arakelov setting. The modular degree is denoted by $\deg(\pi)$.

The APG metric is fixed as the canonical bounded Green-renormalized metric from APG XI:

$$g_{\text{APG}} = e^{2\varphi_{\text{APG}}} g_{\text{std}}, \tag{4}$$

where g_{std} is a compact Arakelov-compatible reference metric and φ_{APG} is the bounded Green-renormalized APG conformal factor. The positive APG Laplacian is denoted by Δ_{APG} , and its first positive eigenvalue is

$$\lambda_1(\Delta_{\text{APG}}) > 0. \tag{5}$$

The APG source density is ρ_{APG} , and its mean-zero fluctuation is

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)}. \tag{6}$$

Thus

$$\int_X F_{\text{APG}} d\mu_{\text{APG}} = 0. \quad (7)$$

The APG entropy of a finite weight vector $W = (w_i)_i$ is

$$H(W) = - \sum_i w_i \log w_i. \quad (8)$$

For the binary Euclidean APG weights

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}, \quad (9)$$

one has

$$H(W) = -(w_a \log w_a + w_b \log w_b), \quad 0 \leq H(W) \leq \log 2. \quad (10)$$

This is the same entropy scale that appears in the first-order closure-obstruction theorem of APG I and the stabilized information-geometric formulation of APG II ([Akhtar, 2026a,b](#)).

4 The APG IX–XIII Logical Chain

The APG IX–XIII route consists of five steps.

1. APG IX proves source concentration:

$$\|F_{\text{APG}}\|_{L^2(X)}^2 \leq C_1 H(W)^2. \quad (11)$$

2. APG X combines APG VIII and APG IX to obtain entropy-controlled spectral energy:

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2. \quad (12)$$

3. APG XI proves polynomial spectral-gap stability for the canonical APG metric:

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A}, \quad (13)$$

for constants $c > 0$ and $A \geq 0$.

4. APG XII supplies the conductor/modular-degree dominance input:

$$H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N. \quad (14)$$

5. APG XIII combines (11), (13), and (14) to prove the final spectral-conductor theorem (1).

The proof is short, but its importance lies in the separation of roles. APG IX removes source blow-up. APG XI prevents spectral collapse. APG XII converts entropy-spectral size into conductor/modular-degree size.

5 Analytic Background: The Spectral–Green Bridge

The APG Dirichlet energy is defined by the mean-zero APG potential u_{APG} solving the Poisson equation

$$\Delta_{\text{APG}} u_{\text{APG}} = F_{\text{APG}}, \quad \int_X u_{\text{APG}} d\mu_{\text{APG}} = 0. \quad (15)$$

The corresponding Dirichlet energy is

$$E_{\text{APG}} = \int_X |\nabla_{\text{APG}} u_{\text{APG}}|^2 d\mu_{\text{APG}} = \int_X u_{\text{APG}} \Delta_{\text{APG}} u_{\text{APG}} d\mu_{\text{APG}}. \quad (16)$$

The sign convention is the positive Laplacian convention, consistent with spectral geometry (Chavel, 1984; Berger, 2003).

Theorem 5.1 (Spectral–Green Bridge, APG VIII). *Let X be compact and let Δ_{APG} be the positive APG Laplacian attached to a fixed smooth compact Arakelov-compatible APG metric. If u_{APG} solves (15), then*

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \|F_{\text{APG}}\|_{L^2(X)}^2. \quad (17)$$

Proof. Since u_{APG} has mean zero, the Rayleigh quotient characterization of the first positive eigenvalue gives

$$\lambda_1(\Delta_{\text{APG}}) \|u_{\text{APG}}\|_{L^2(X)}^2 \leq \int_X |\nabla_{\text{APG}} u_{\text{APG}}|^2 d\mu_{\text{APG}} = E_{\text{APG}}. \quad (18)$$

Using $\Delta_{\text{APG}} u_{\text{APG}} = F_{\text{APG}}$, integration by parts gives

$$E_{\text{APG}} = \int_X u_{\text{APG}} F_{\text{APG}} d\mu_{\text{APG}} \leq \|u_{\text{APG}}\|_{L^2(X)} \|F_{\text{APG}}\|_{L^2(X)}. \quad (19)$$

If $E_{\text{APG}} = 0$, the assertion is immediate. Otherwise, (18) gives

$$\|u_{\text{APG}}\|_{L^2(X)} \leq \lambda_1(\Delta_{\text{APG}})^{-1/2} E_{\text{APG}}^{1/2}.$$

Substituting into (19) and dividing by $E_{\text{APG}}^{1/2}$ yields

$$E_{\text{APG}}^{1/2} \leq \lambda_1(\Delta_{\text{APG}})^{-1/2} \|F_{\text{APG}}\|_{L^2(X)}.$$

Squaring proves (17). □

This theorem is not the final APG XIII inequality, but it explains why the object in Open Problem 15.1 is the natural spectral object. The proof uses only standard Hilbert-space spectral theory on compact Riemann surfaces (Chavel, 1984; Berger, 2003).

6 Imported Theorems

This section records the exact inputs used in APG XIII.

Theorem 6.1 (Source Concentration, APG IX). *For the APG source density on the compactified modular curve X , the mean-zero fluctuation F_{APG} belongs to $L^2(X)$ and satisfies*

$$\|F_{\text{APG}}\|_{L^2(X)}^2 \leq C_1 H(W)^2. \quad (20)$$

The estimate includes cusp neighborhoods under the APG cusp-regularized normalization.

Remark 6.2. *The content of APG IX is not merely that ρ_{APG} is formally defined. The theorem proves that the source fluctuation is square-integrable and cannot develop non-integrable cusp concentration. This is the source-side reason the APG spectral-conductor program can proceed.*

Theorem 6.3 (Spectral-Gap Stability, APG XI). *For the canonical bounded Green-renormalized APG metric $g_{\text{APG}} = e^{2\varphi_{\text{APG}}} g_{\text{std}}$, there are constants $c > 0$ and $A \geq 0$ such that*

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A}. \quad (21)$$

Equivalently, for $C_2 = c^{-1}$,

$$\lambda_1(\Delta_{\text{APG}})^{-1} \leq C_2 (\log N)^A. \quad (22)$$

Remark 6.4. *The restriction to the canonical bounded Green-renormalized APG metric is essential. APG XI does not claim spectral stability for arbitrary unregulated conformal metrics. It proves polynomial stability for the metric normalization adopted by APG XII and APG XIII.*

Theorem 6.5 (Conductor–Modular Degree Dominance, APG XII). *Under the canonical APG boundary realization adopted for the APG IX–XIII route, there is a constant $C_3 > 0$ such that*

$$H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N. \quad (23)$$

Remark 6.6. *APG XII obtains the conductor-scale dominance through the boundary package*

$$B_{\text{APG}}(N, \pi) = B_{\text{cusp}}(N, \pi) + B_{\text{vert}}(N, \pi) + B_{\text{Green}}(N, \pi), \quad (24)$$

whose terms measure cusp-width complexity, vertical arithmetic complexity, and Green-energy metric distortion. Its role in APG XIII is exactly (23).

7 Main Theorem: Resolution of Open Problem 15.1

Theorem 7.1 (APG XIII Spectral–Conductor Theorem). *Assume the APG IX source concentration theorem, the APG XI canonical spectral-gap theorem, and the APG XII conductor–modular degree dominance theorem. Then there exists a constant $C > 0$, independent of the APG source fluctuation and depending only on the canonical constants in the preceding estimates, such that*

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C \deg(\pi) \log N. \quad (25)$$

Consequently, Open Problem 15.1 is resolved within the canonical APG IX–XIII framework.

Proof. By definition $F_{\text{APG}} = \rho_{\text{APG}} - 1/\text{Vol}(X)$. APG IX gives

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 = \|F_{\text{APG}}\|_{L^2(X)}^2 \leq C_1 H(W)^2. \quad (26)$$

APG XI gives

$$\lambda_1(\Delta_{\text{APG}})^{-1} \leq C_2 (\log N)^A. \quad (27)$$

Multiplying (26) and (27), we obtain

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 C_2 H(W)^2 (\log N)^A. \quad (28)$$

APG XII gives

$$H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N. \quad (29)$$

Substituting (29) into (28) yields

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 C_2 C_3 \deg(\pi) \log N. \quad (30)$$

Set

$$C = C_1 C_2 C_3. \quad (31)$$

Then (30) becomes exactly (25). This proves the theorem. $\square$

Corollary 7.2 (Canonical Spectral Source Control). *The spectral size of the APG mean-zero source is conductor-controlled:*

$$\|F_{\text{APG}}\|_{H_{\text{APG}}^{-1}(X)}^2 \leq C \deg(\pi) \log N, \quad (32)$$

where the negative Sobolev norm is defined spectrally by

$$\|F\|_{H_{\text{APG}}^{-1}(X)}^2 = \sum_{j \geq 1} \frac{|\langle F, e_j \rangle|^2}{\lambda_j(\Delta_{\text{APG}})} \quad (33)$$

for an orthonormal eigenbasis $(e_j)_{j \geq 0}$ of $L^2(X)$, with e_0 constant.

Proof. Since F is mean zero,

$$\|F\|_{H_{\text{APG}}^{-1}(X)}^2 \leq \lambda_1(\Delta_{\text{APG}})^{-1} \|F\|_{L^2(X)}^2.$$

Apply Theorem 7.1 to $F = F_{\text{APG}}$. $\square$

Corollary 7.3 (APG Energy Bound). *The APG Dirichlet energy satisfies*

$$E_{\text{APG}} \leq C \deg(\pi) \log N. \quad (34)$$

Proof. The Spectral–Green Bridge (17) gives

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \|F_{\text{APG}}\|_{L^2(X)}^2.$$

Theorem 7.1 bounds the right-hand side by $C \deg(\pi) \log N$. $\square$

8 Constant Bookkeeping

The constants in the proof have distinct origins:

$$C_1 \quad \text{from APG IX source concentration,} \tag{35}$$

$$C_2 = c^{-1} \quad \text{from APG XI spectral-gap inversion,} \tag{36}$$

$$C_3 \quad \text{from APG XII conductor–modular degree dominance.} \tag{37}$$

Thus the final APG XIII constant is

$$C = C_1 C_2 C_3. \tag{38}$$

No step in the proof requires the energy bound from APG X directly. APG X is nevertheless structurally important because it explains why source concentration and spectral gap are the two analytic factors controlling APG energy.

9 Asymptotic Interpretation of the Spectral–Conductor Bound

The proof of Theorem 7.1 is entirely symbolic and does not depend on any particular numerical value of the conductor N . Consequently, the APG XIII spectral–conductor theorem is most naturally understood through its asymptotic scaling behavior rather than through a specific numerical example.

The principal quantity appearing in Open Problem 15.1 is

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2. \tag{39}$$

By the APG IX source concentration theorem,

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 H(W)^2. \tag{40}$$

The APG XI spectral-gap theorem yields

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A}, \tag{41}$$

and therefore

$$\lambda_1(\Delta_{\text{APG}})^{-1} \leq c^{-1} (\log N)^A. \tag{42}$$

Combining (40) and (42) yields

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 c^{-1} H(W)^2 (\log N)^A. \quad (43)$$

Thus the analytic contribution of the APG source is amplified by at most a polylogarithmic factor in the conductor.

The APG XII conductor–modular degree dominance theorem asserts that

$$H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N. \quad (44)$$

Substituting (44) into (43) gives

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 c^{-1} C_3 \deg(\pi) \log N. \quad (45)$$

Consequently,

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 = \mathcal{O}(\deg(\pi) \log N). \quad (46)$$

Equation (46) provides the conceptual interpretation of the APG XIII spectral–conductor theorem. The source concentration mechanism of APG IX contributes the entropy scale $H(W)^2$, APG XI permits only polylogarithmic amplification through the factor $(\log N)^A$, and APG XII demonstrates that this amplified entropy scale is ultimately dominated by the arithmetic growth quantity $\deg(\pi) \log N$.

Therefore, APG XIII establishes that APG spectral source complexity cannot grow faster than conductor–modular degree growth. The theorem should be interpreted as a conductor-scale control principle for APG spectral fluctuations rather than as a numerical estimate attached to any specific modular level.

10 Conditional Fermat Obstruction Consequence

The APG XIII theorem is an upper bound. By itself it does not prove Fermat’s Last Theorem. However, it gives a clean conditional obstruction principle.

Definition 10.1 (Strict APG lower-gap condition). *A primitive Fermat triple (a, b, c, p) , if it existed, is said to satisfy the strict APG lower-gap condition if its associated APG energy obeys*

$$E_{\text{APG}}(a, b, c, p) \geq C' \deg(\pi) \log N \quad (47)$$

for some constant $C' > C$, where C is the APG XIII constant in Theorem 7.1 and Corollary 7.3.

Theorem 10.2 (Conditional Fermat Obstruction). *Assume the APG XIII spectral–conductor theorem and suppose every primitive Fermat triple*

$$a^p + b^p = c^p, \quad p > 2, \quad (48)$$

forces the strict APG lower-gap condition (47). Then no primitive Fermat triple exists.

Proof. Suppose, for contradiction, that a primitive Fermat triple exists. By the strict APG lower-gap condition,

$$E_{\text{APG}} \geq C' \deg(\pi) \log N. \quad (49)$$

By Corollary 7.3, APG XIII gives

$$E_{\text{APG}} \leq C \deg(\pi) \log N. \quad (50)$$

Combining (49) and (50),

$$C' \deg(\pi) \log N \leq C \deg(\pi) \log N. \quad (51)$$

Since $\deg(\pi) > 0$ and $\log N > 0$, division gives

$$C' \leq C. \quad (52)$$

This contradicts $C' > C$. Therefore no primitive Fermat triple satisfying the strict APG lower-gap condition exists. If every primitive Fermat triple satisfies that condition, then no primitive Fermat triple exists. $\square$

Remark 10.3. *Theorem 10.2 is a conditional obstruction theorem, not an unconditional new proof of Fermat's Last Theorem. Its value is that it identifies the exact remaining APG-to-Fermat condition: a strict lower energy gap for hypothetical Fermat triples. The classical proof of Fermat's Last Theorem remains the modular proof based on Frey curves, Mazur's work on modular curves and rational isogenies, Ribet's level-lowering theorem, and Wiles–Taylor–Wiles modularity lifting (Frey, 1986; Mazur, 1977, 1978; Ribet, 1990; Taylor and Wiles, 1995; Wiles, 1995).*

11 Discussion

The proof of Open Problem 15.1 is short because the difficult work is distributed across APG IX, APG XI, and APG XII. This is a strength rather than a weakness. APG XIII has one job: to show that the previously isolated analytic and arithmetic estimates combine without circularity.

The most delicate input remains APG XI. If spectral-gap stability failed for the APG metric, then the inverse spectral factor could dominate the source bound and the proof would collapse. APG XI avoids this by adopting the canonical bounded Green-renormalized metric and proving conductor-polynomial spectral stability. The second delicate input is APG XII. It must ensure that the entropy scale remaining after APG IX and APG XI is absorbed by $\deg(\pi) \log N$. Once these two estimates are in place, the APG XIII proof is purely formal.

The final theorem should therefore be read as a spectral-conductor synthesis, not as a new modularity theorem. It does not replace Frey–Ribet–Wiles. It establishes that the APG energy/source architecture, after canonical metric normalization and conductor dominance, satisfies the conductor-scale spectral inequality demanded by Open Problem 15.1.

12 Future Directions and the APG–Fermat Program

The APG XIII spectral–conductor theorem completes the canonical APG IX–XIII route associated with Open Problem 15.1. The theorem establishes that APG spectral source complexity is controlled by conductor and modular-degree growth through the inequality

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right|_{L^2(X)}^2 \leq C, \deg(\pi), \log N. \quad (53)$$

This result should be viewed as a conductor-scale control theorem rather than as the final objective of the APG program. Its significance is that it identifies a universal upper bound for APG spectral complexity and demonstrates that entropy-controlled source fluctuations remain dominated by arithmetic conductor growth after spectral amplification.

Within the broader APG research program, one long-term motivation has been to investigate whether deformation-theoretic, entropy-based, and spectral methods can provide new perspectives on classical Diophantine problems. Earlier APG volumes explored conditional links between APG invariants, Frey-type elliptic curves, modular parametrizations, arithmetic heights, and Arakelov-theoretic structures. In that setting, Fermat’s Last Theorem serves not as a solved target within APG, but as a benchmark problem against which the strength of the APG framework can be measured.

The present theorem does not provide a new proof of Fermat’s Last Theorem. Instead, it identifies the precise location at which any future APG-based obstruction argument must occur. The APG XIII theorem supplies a conductor-scale upper bound. Consequently, any contradiction argument must arise from a sufficiently strong lower-bound mechanism.

More specifically, if one could establish that a hypothetical primitive Fermat triple necessarily forces an APG energy growth condition of the form

$$E_{\text{APG}} \geq C', \deg(\pi), \log N, \quad C' > C, \quad (54)$$

then the APG XIII conductor wall would immediately yield a contradiction. The central open problem for the APG–Fermat program is therefore the construction of a rigorous lower-gap theorem connecting hypothetical Fermat solutions to unavoidable APG spectral, geometric, or entropy-driven complexity.

More broadly, future investigations may explore whether entropy-controlled deformation invariants, spectral concentration phenomena, modular parametrizations, arithmetic heights, and conductor-scale growth principles can provide new insights into arithmetic geometry and Diophantine obstruction theory. The APG framework remains under active development, and the present theorem identifies a precise boundary between what has been established and what remains to be understood.

13 Conclusion

APG XIII proves the final spectral-conductor inequality

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C \deg(\pi) \log N.$$

The proof combines three earlier APG milestones:

1. APG IX: $\|F_{\text{APG}}\|_2^2 \leq C_1 H(W)^2$;
2. APG XI: $\lambda_1(\Delta_{\text{APG}})^{-1} \leq C_2 (\log N)^A$;
3. APG XII: $H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N$.

Multiplication of the first two estimates and substitution of the third gives the result with $C = C_1 C_2 C_3$. The manuscript also records the conditional Fermat obstruction consequence: APG XIII plus a strict APG lower-energy gap for primitive Fermat triples would rule out such triples. That final lower-gap assertion is not claimed here as unconditional.

Acknowledgment of Scope

This manuscript is written as the final synthesis volume of the APG IX–XIII spectral-conductor route. It proves Open Problem 15.1 within the stated canonical APG hypotheses. It does not claim to prove Fermat’s Last Theorem independently of the classical modular method, nor does it claim proofs of the *abc*, Szpiro, or Birch and Swinnerton-Dyer conjectures.

About the Author and the APG Program

Dr. Mohammad Amir Khusru Akhtar is engaged in research spanning mathematical modeling, information theory, geometric methods, data analysis, artificial intelligence, and interdisciplinary applications of science and technology. His work seeks to uncover structural relationships between geometry, information, computation, and arithmetic through the development of new mathematical frameworks and analytical methods.

A central focus of his current mathematical research is the development of *Arithmetic Power Geometry (APG)*, an ongoing research program whose conceptual origins trace back to ideas first explored in the early 1990s. The APG framework investigates exponent-deformation geometry, entropy-controlled invariants, information-geometric structures, spectral methods, arithmetic regularization, and their interactions with modular and arithmetic phenomena.

The long-term objective of the APG program is to explore whether geometric deformation, information, entropy, and arithmetic structure can be understood within a common mathematical language. While many aspects of the theory remain under active development, the program aims to provide new perspectives on the interplay between geometry, analysis, information, and number theory.

The present APG XIII manuscript forms part of this continuing effort. By establishing the spectral-conductor theorem associated with Open Problem 15.1, it contributes to the broader

goal of understanding how entropy, spectral structure, and arithmetic growth interact within the APG framework.

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